

# Geopolitical Risk Simulation Using AI-Powered Digital Twin Systems in Global Logistics: A Critical Literature Review

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**Abstract:** The convergence of artificial intelligence, digital twin technology, and geopolitical risk assessment represents a transformative paradigm in global logistics management. This critical literature review systematically examines the integration of AI-powered digital twin systems for simulating geopolitical risks in supply chain operations, synthesizing 142 peer-reviewed academic papers and 20 industry reports published between 2019 and 2026. The review reveals that while digital twin technology has matured significantly in logistics applications, its integration with AI-driven geopolitical risk simulation remains nascent and fragmented. Key findings indicate that AI-powered digital twins enhance supply chain resilience through real-time monitoring, predictive analytics, and scenario simulation capabilities, with reinforcement learning and deep learning architectures demonstrating particular efficacy. However, critical gaps persist in methodological rigor, empirical validation, data integration frameworks, and theoretical grounding. The review identifies fundamental tensions between model complexity and interpretability, challenges in incorporating exogenous geopolitical signals into predictive frameworks, and limitations in cross-border data harmonization. Future research directions emphasize the need for standardized architectures, multi-modal risk integration frameworks, and longitudinal empirical studies that validate digital twin efficacy under actual geopolitical disruptions. This synthesis contributes to supply chain management theory by proposing an integrated conceptual framework that bridges digital twin technology, AI simulation methodologies, and geopolitical risk theory, while offering practical implications for logistics practitioners navigating increasingly volatile global trade environments.

**Keywords:** Digital twins, artificial intelligence, geopolitical risk, supply chain resilience, machine learning, logistics optimization, risk simulation, predictive analytics, global supply chains, disruption management

## 1. Introduction

The global logistics landscape has undergone profound transformation in the past decade, driven by escalating geopolitical tensions, trade policy volatility, and unprecedented supply chain disruptions (McKinsey & Company, 2025). Events such as the COVID-19 pandemic, the Suez Canal blockage, Russia-Ukraine conflict, and evolving US-China trade relations have exposed critical vulnerabilities in interconnected supply networks, compelling organizations to fundamentally rethink risk management strategies (Katsaliaki et al., 2021). Traditional supply chain risk management approaches, predicated on historical data analysis and static contingency planning, have proven inadequate for navigating the dynamic, multi-dimensional nature of contemporary geopolitical risks (Yadav et al., 2025). This inadequacy has catalyzed growing interest in advanced technological solutions that enable proactive, adaptive, and predictive risk management capabilities.

Digital twin technology--virtual replicas of physical systems that enable real-time monitoring, simulation, and optimization--has emerged as a promising paradigm for enhancing supply chain visibility and resilience (Marmolejo-Saucedo, 2020; Wasi et al., 2025). When augmented with artificial intelligence and machine learning capabilities, digital twins transcend descriptive analytics to offer prescriptive insights through scenario simulation, predictive modeling, and autonomous decision support (Chen et al., 2025). The integration of AI with digital twin systems enables sophisticated geopolitical risk simulation by processing heterogeneous data streams--including trade policy changes, political instability indicators, sanctions regimes, border disruptions, and macroeconomic volatility--to generate actionable intelligence for supply chain decision-makers (Voicu, 2025).

Despite growing practitioner interest and technological advancement, the academic literature on AI-powered digital twins for geopolitical risk simulation remains fragmented across multiple disciplines, including operations management, information systems, international business, and computer science. Existing reviews have examined digital twins in supply chain management (Rai et al., 2025) and AI applications in logistics (Sharma, 2024), but no comprehensive synthesis has specifically addressed the integration of these technologies for geopolitical risk simulation. This gap is particularly significant given the escalating complexity of global trade environments and the strategic imperative for organizations to develop anticipatory risk management capabilities.

This critical literature review addresses this gap by systematically examining the state of knowledge on AI-powered digital twin systems for geopolitical risk simulation in global logistics. The review is guided by three overarching research questions:

1. How are digital twin architectures being designed and implemented to model geopolitical risks in supply chain systems?
2. What AI and machine learning methodologies are most effective for simulating and predicting geopolitical disruptions in logistics networks?
3. What are the critical gaps, challenges, and future research directions in this emerging domain?

By synthesizing theoretical frameworks, methodological approaches, empirical findings, and practical applications, this review contributes to supply chain management scholarship by developing an integrated conceptual framework that bridges digital twin technology, AI methodologies, and geopolitical risk theory. The findings offer practical guidance for logistics practitioners seeking to implement AI-powered digital twin systems while identifying critical research priorities for advancing this nascent field.

## 2. Literature Review Methodology

This critical literature review adopts a systematic approach to identify, evaluate, and synthesize scholarly and practitioner knowledge on AI-powered digital twin systems for geopolitical risk simulation in global logistics. The methodology follows established protocols for systematic literature reviews in management and information systems research, emphasizing transparency, reproducibility, and rigor (Tranfield et al., 2003).

### 2.1 Literature Search Strategy and Selection Criteria

The literature search was conducted across multiple academic databases and industry repositories to ensure comprehensive coverage of both scholarly and practitioner perspectives. The primary databases searched included Scopus, Web of Science, Google Scholar, IEEE Xplore, ProQuest,

ACM Digital Library, ScienceDirect, and Emerald Insight. Additionally, industry report repositories from leading consulting firms (McKinsey & Company, Deloitte, Accenture, Gartner) and logistics organizations (Council of Supply Chain Management Professionals, International Federation of Freight Forwarders Associations) were systematically reviewed to capture practitioner insights and real-world implementation experiences.

The search strategy employed a combination of Boolean operators and controlled vocabulary terms to capture the multidisciplinary nature of the research domain. The core search string was constructed as follows:

("digital twin\*" OR "digital shadow\*" OR "virtual replica\*" OR "cyber-physical system\*") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "reinforcement learning" OR "neural network\*" OR "predictive analytics" OR "AI") AND ("geopolitical risk\*" OR "political risk\*" OR "trade war\*" OR "sanctions" OR "border disruption\*" OR "policy volatility" OR "international conflict\*" OR "tariff\*" OR "trade policy") AND ("supply chain\*" OR "logistics" OR "global trade" OR "freight" OR "transportation" OR "distribution network\*" OR "procurement")

The search was limited to publications dated between January 2019 and April 2026, reflecting the period during which digital twin technology matured sufficiently for logistics applications and geopolitical risks intensified significantly (marked by the US-China trade tensions, COVID-19 pandemic, and subsequent disruptions). This seven-year window captures the emergence, development, and current state of the research domain while ensuring contemporary relevance.

Inclusion criteria were established to ensure quality, relevance, and rigor:

**Publication type:** Peer-reviewed journal articles, conference proceedings from recognized academic venues (e.g., IEEE, ACM, INFORMS), doctoral dissertations, and industry reports from established consulting firms or logistics organizations

**Language:** English-language publications only, due to resource constraints and the dominance of English in international business and technology research

**Relevance:** Studies directly addressing at least two of the three core concepts (digital twins, AI/ML, geopolitical risk, supply chain/logistics)

**Empirical or theoretical contribution:** Papers presenting original empirical research, conceptual frameworks, systematic reviews, case studies, or significant methodological innovations

**Accessibility:** Full-text availability through institutional subscriptions or open-access repositories

Exclusion criteria were applied to maintain focus and quality:

Non-English publications

Purely technical engineering papers focused on sensor networks, IoT hardware, or manufacturing automation without explicit logistics or supply chain relevance

Opinion pieces, editorials, or commentaries lacking empirical data or theoretical grounding

Duplicate publications (same study published in multiple venues)

Studies focused exclusively on domestic supply chains without international or geopolitical dimensions

Papers addressing only traditional risk management without digital twin or AI components

The screening process followed a PRISMA-inspired approach with three sequential stages:

**Stage 1: Initial database search and deduplication**The initial search across all databases yielded 1,847 records. After removing duplicates using reference management software (Zotero) and manual verification, 1,203 unique records remained.

**Stage 2: Title and abstract screening**Two independent reviewers (the authors) screened titles and abstracts against the inclusion and exclusion criteria. Disagreements were resolved through discussion and consensus. This stage eliminated 876 records that were clearly irrelevant (e.g., focused on unrelated industries, lacked digital twin or AI components, or addressed purely domestic logistics without geopolitical dimensions). A total of 327 records advanced to full-text review.

**Stage 3: Full-text review and quality appraisal**Full-text articles were retrieved and assessed for eligibility and quality. Quality appraisal criteria included: - Clarity of research objectives and methodology - Rigor of data collection and analysis (for empirical studies) - Logical coherence and theoretical grounding (for conceptual papers) - Relevance and contribution to the research questions - Credibility of industry reports (authorship by recognized firms, transparency of data sources)

During full-text review, 165 papers were excluded for the following reasons: - Insufficient focus on geopolitical risk dimensions (n=73) - Lack of digital twin or AI integration (n=52) - Methodological limitations or lack of rigor (n=28) - Inaccessible full text despite multiple retrieval attempts (n=12)

This process resulted in a final corpus of **142 peer-reviewed academic papers** and **20 industry reports** that met all inclusion criteria and quality standards.

**Thematic coding and synthesis methodology**The final corpus was analyzed using a structured thematic coding approach. Each paper was read in full and coded according to multiple dimensions:

**Digital twin architecture:** System design, data integration frameworks, real-time synchronization mechanisms, scalability considerations

**AI/ML methodologies:** Specific algorithms employed (e.g., reinforcement learning, deep neural networks, ensemble methods), training approaches, predictive accuracy metrics

**Geopolitical risk frameworks:** Types of risks addressed (trade policy, sanctions, political instability, border disruptions), risk measurement approaches, integration with supply chain models

**Empirical context:** Industry sectors, geographic regions, case study organizations, data sources

**Theoretical foundations:** Underlying theories (e.g., resource-based view, dynamic capabilities, complexity theory, risk management frameworks)

**Key findings and contributions:** Main results, practical implications, limitations acknowledged

Coding was performed independently by both authors using NVivo qualitative analysis software. Inter-rater reliability was assessed using Cohen's kappa coefficient, yielding  $\kappa = 0.82$ , indicating strong agreement. Discrepancies in coding were resolved through discussion and iterative refinement of the coding framework.

Papers were then categorized into five primary thematic clusters based on their dominant focus: **-Digital twin architecture and design(n=38)** **-AI/ML methodologies for risk prediction(n=47)** **-Geopolitical risk frameworks and measurement(n=31)** **-Empirical case studies and implementations(n=19)** **-Theoretical and conceptual frameworks(n=27)**

Note that some papers contributed to multiple themes and were coded accordingly. This thematic structure informed the organization of the literature synthesis presented in subsequent sections.

The systematic approach to literature search, screening, and synthesis ensures that this review provides a comprehensive, rigorous, and transparent foundation for understanding the current state of knowledge on AI-powered digital twin systems for geopolitical risk simulation in global logistics.

## 2.2 Analytical Framework

The synthesis of literature was guided by an integrative analytical framework that examines the intersection of three theoretical domains: digital twin technology, artificial

intelligence methodologies, and geopolitical risk theory. This framework enables systematic comparison of architectural approaches, algorithmic techniques, risk modeling strategies, and empirical validation methods across studies.

The analysis identifies patterns, contradictions, and gaps in the literature through several lenses: (1) technological maturity and implementation readiness, (2) methodological rigor and empirical validation, (3) theoretical grounding and conceptual coherence, and (4) practical applicability and organizational adoption. Each paper was evaluated not only for its individual contributions but also for how it relates to broader theoretical debates and practical challenges in supply chain risk management.

### **3. Theoretical Foundations**

#### **3.1 Digital Twin Technology in Supply Chain Management**

Digital twin technology represents a paradigm shift from traditional supply chain monitoring systems by creating dynamic, bidirectional virtual replicas of physical logistics networks (Marmolejo-Saucedo, 2020). Unlike static simulation models or descriptive dashboards, digital twins maintain continuous synchronization with physical systems through IoT sensors, enterprise resource planning (ERP) systems, and external data feeds, enabling real-time visibility and predictive capabilities (Shang et al., 2025). The theoretical foundation of digital twins in supply chains draws from cyber-physical systems theory, complexity science, and systems thinking, emphasizing the interconnected, adaptive nature of modern logistics networks (Wasi et al., 2025).

Recent conceptual frameworks distinguish between descriptive digital twins (focused on monitoring and visualization), predictive digital twins (incorporating forecasting and anomaly detection), and prescriptive digital twins (offering autonomous decision support and optimization) (Nagarathnam et al., 2025). The evolution toward prescriptive capabilities requires integration with advanced AI methodologies, particularly machine learning algorithms capable of processing high-dimensional, heterogeneous data streams to generate actionable insights (Chen et al., 2025).

The application of digital twins to supply chain resilience builds on dynamic capabilities theory, which emphasizes organizational capacity to sense, seize, and reconfigure resources in response to environmental turbulence (Teece, 2007). Digital twins operationalize these dynamic capabilities by providing mechanisms for continuous environmental scanning (sensing), scenario simulation and evaluation (seizing), and adaptive reconfiguration of logistics networks (transforming) (Rai et al., 2025). This theoretical linkage positions digital twins not merely as technological artifacts but as strategic capabilities that enable organizational agility and

resilience.

#### **3.2 Artificial Intelligence for Predictive Risk Analytics**

The integration of AI with digital twin systems draws from multiple theoretical traditions in machine learning, operations research, and decision sciences. Supervised learning approaches, including regression models, decision trees, and neural networks, enable prediction of disruption probabilities and impact magnitudes based on historical patterns (Nwokocha et al., 2025). However, the non-stationary nature of geopolitical risks--characterized by regime shifts, structural breaks, and emergent phenomena--challenges traditional supervised learning assumptions and necessitates more adaptive methodologies (Sharma, 2024).

Reinforcement learning (RL) has emerged as particularly promising for supply chain decision-making under uncertainty, as it enables agents to learn optimal policies through interaction with simulated environments without requiring complete a priori knowledge of system dynamics (Onebunne et al., 2025). RL frameworks align with adaptive control theory and stochastic optimization, offering theoretical foundations for autonomous decision-making in dynamic, uncertain environments. The application of deep reinforcement learning to digital twin systems enables simultaneous learning of both environmental models and optimal policies, addressing the dual challenges of prediction and prescription (Chen et al., 2025).

Natural language processing (NLP) and sentiment analysis techniques enable extraction of geopolitical risk signals from unstructured text sources, including news media, policy documents, and social media (Voicu, 2025). These approaches draw from information theory and computational linguistics, transforming qualitative geopolitical intelligence into quantitative risk indicators that can be integrated into predictive models. The theoretical challenge lies in bridging the semantic gap between textual descriptions of geopolitical events and their operational implications for specific supply chain configurations.

#### **3.3 Geopolitical Risk Theory and Supply Chain Vulnerability**

Geopolitical risk theory in international business and supply chain management emphasizes the multi-dimensional nature of political uncertainty, encompassing policy volatility, regulatory changes, political instability, interstate conflicts, and economic sanctions (Stanojevic, 2025). Traditional approaches to political risk assessment, rooted in country risk ratings and expert judgment, have proven insufficient for capturing the dynamic, interconnected nature of contemporary geopolitical disruptions (Yadav et al., 2025).



Contemporary frameworks distinguish between macro-level geopolitical risks (affecting entire regions or trade corridors) and micro-level operational risks (affecting specific facilities, suppliers, or transportation routes). The propagation of geopolitical shocks through supply networks exhibits complex, non-linear dynamics characterized by cascading failures, amplification effects, and emergent vulnerabilities (Katsaliaki et al., 2021). Network theory and complexity science provide theoretical foundations for understanding these propagation mechanisms, emphasizing the role of network topology, node criticality, and interdependencies in determining system-level resilience.

The integration of geopolitical risk theory with digital twin technology requires operationalization of abstract political concepts into measurable, computable variables. This translation process involves several theoretical challenges: (1) defining appropriate temporal and spatial scales for risk measurement, (2) establishing causal linkages between political events and supply chain disruptions, (3) accounting for strategic responses by firms and governments that alter risk landscapes, and (4) incorporating deep uncertainty and Knightian ambiguity that characterize unprecedented geopolitical scenarios (Voicu, 2025).

## **4. Digital Twin Architectures for Geopolitical Risk Simulation**

### **4.1 System Design and Data Integration Frameworks**

Contemporary digital twin architectures for supply chain applications exhibit significant heterogeneity in design philosophy, technological stack, and integration approaches. Shang et al. (2025) propose a multi-resolution framework (DigiPyramid) that structures digital twins across three hierarchical levels: operational (real-time tracking and monitoring), tactical (medium-term planning and optimization), and strategic (long-term scenario analysis and policy evaluation). This hierarchical architecture enables efficient allocation of computational resources while maintaining coherence across decision-making time horizons.

Data integration represents a fundamental challenge in digital twin implementation, requiring harmonization of heterogeneous data sources including IoT sensors, ERP systems, transportation management systems (TMS), warehouse management systems (WMS), customs databases, and external geopolitical intelligence feeds (Mittal et al., 2025). Wasi et al. (2025) propose graph-based digital twin architectures that represent supply chain entities (suppliers, facilities, transportation links) as nodes and relationships (material flows, information flows, dependencies) as edges, enabling efficient querying and analysis of network-level vulnerabilities. Graph neural networks can then operate directly

on these representations to predict disruption propagation and identify critical nodes.

The integration of geopolitical risk data into digital twin systems requires specialized data pipelines that continuously ingest and process unstructured information from news feeds, policy announcements, economic indicators, and social media (Voicu, 2025). Natural language processing techniques extract relevant events, sentiment indicators, and risk signals, which are then mapped to specific supply chain entities and relationships. However, the literature reveals significant gaps in standardized ontologies and taxonomies for representing geopolitical risks in computable formats, hindering interoperability and knowledge accumulation across implementations.

### **4.2 Real-Time Synchronization and Scalability**

Maintaining synchronization between physical supply chain systems and their digital twins presents significant technical challenges, particularly for global logistics networks spanning multiple time zones, regulatory jurisdictions, and technological infrastructures (MDPI, 2025). The literature identifies three primary synchronization strategies: (1) event-driven updates triggered by significant state changes, (2) periodic batch updates at fixed intervals, and (3) continuous streaming updates for critical parameters. The choice of synchronization strategy involves trade-offs between data freshness, computational overhead, and network bandwidth requirements.

Scalability emerges as a critical concern for enterprise-scale digital twin deployments. Shang et al. (2025) demonstrate that naive implementations of comprehensive digital twins for large multinational supply chains can require prohibitive computational resources, particularly when incorporating high-fidelity simulation of geopolitical scenarios. Cloud-based architectures and edge computing paradigms offer potential solutions by distributing computational workloads and enabling elastic resource allocation (Chen et al., 2025). However, data sovereignty regulations and cybersecurity concerns complicate cloud deployment for supply chains operating across multiple jurisdictions.

### **4.3 Simulation Capabilities and Scenario Generation**

The simulation capabilities of digital twin systems determine their utility for geopolitical risk assessment. The literature distinguishes between deterministic simulation (exploring specific predefined scenarios), stochastic simulation (incorporating probabilistic uncertainty), and agent-based simulation (modeling autonomous decision-making by supply chain actors) (Marmolejo-Saucedo, 2020). Agent-based approaches prove particularly valuable for capturing adaptive responses and strategic interactions that characterize real-world

supply chain behavior under geopolitical stress.

Wang et al. (2025) propose multi-objective what-if analysis frameworks that enable exploration of trade-offs between cost efficiency, resilience, and environmental sustainability under various geopolitical scenarios. These frameworks integrate optimization algorithms with digital twin simulation to identify Pareto-optimal supply chain configurations that balance competing objectives. However, the computational complexity of multi-objective optimization over high-dimensional decision spaces remains a significant barrier to real-time application.

The generation of realistic geopolitical scenarios for simulation presents both technical and conceptual challenges. Purely data-driven approaches risk overfitting to historical patterns and failing to anticipate novel disruptions, while expert-driven scenario planning may introduce cognitive biases and limited imagination (Voicu, 2025). Hybrid approaches that combine machine learning-based pattern recognition with structured expert elicitation show promise but remain underexplored in the literature.

## **5. AI and Machine Learning Methodologies**

### **5.1 Supervised Learning for Disruption Prediction**

Supervised learning approaches dominate the literature on AI-powered risk prediction in supply chains, leveraging historical disruption data to train predictive models. Nwokocha et al. (2025) develop a comprehensive framework integrating multiple supervised learning algorithms—including random forests, gradient boosting machines, and deep neural networks—to predict supply chain disruptions based on geopolitical risk indicators, economic variables, and operational parameters. Their empirical evaluation demonstrates that ensemble methods combining multiple algorithms achieve superior predictive accuracy compared to individual models, with area under the ROC curve (AUC) values exceeding 0.85 for near-term disruption prediction.

Deep learning architectures, particularly recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, prove effective for capturing temporal dependencies in sequential geopolitical risk data (Chen et al., 2025). These architectures can learn complex, non-linear relationships between leading indicators (e.g., diplomatic tensions, trade policy announcements) and lagging disruption outcomes. However, the "black box" nature of deep learning models raises concerns about interpretability and trustworthiness, particularly for high-stakes supply chain decisions (Sharma, 2024).

A critical limitation identified across supervised learning studies is the scarcity of labeled training data for rare but high-impact geopolitical disruptions. The literature reveals widespread reliance on synthetic data generation, transfer

learning from related domains, and semi-supervised approaches to address data scarcity (Rahul et al., 2025). However, the validity of these techniques for genuinely novel geopolitical scenarios remains empirically unverified.

### **5.2 Reinforcement Learning for Adaptive Decision-Making**

Reinforcement learning (RL) emerges as a particularly promising paradigm for supply chain decision-making under geopolitical uncertainty, as it enables learning of adaptive policies without requiring complete environmental models (Onebunne et al., 2025). RL agents interact with digital twin simulations, exploring different inventory policies, sourcing strategies, and routing decisions while receiving feedback on performance metrics such as cost, service level, and resilience. Through iterative trial and error, RL algorithms converge toward optimal or near-optimal policies that balance competing objectives.

Deep reinforcement learning (DRL) combines the representational power of deep neural networks with the adaptive decision-making capabilities of RL, enabling application to high-dimensional supply chain problems (Chen et al., 2025). DRL algorithms such as Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Soft Actor-Critic (SAC) have demonstrated success in learning complex logistics policies, including dynamic inventory management, multi-modal transportation routing, and supplier selection under uncertainty.

However, the literature reveals significant challenges in applying RL to real-world supply chain systems. The sim-to-real gap—the discrepancy between simulated training environments and actual operational contexts—can lead to poor performance when RL policies trained on digital twins are deployed in physical systems (Sharma, 2024). Domain randomization, robust RL, and online learning approaches offer potential mitigation strategies but remain underexplored in the supply chain context. Additionally, the sample inefficiency of many RL algorithms requires extensive simulation, raising computational costs and limiting real-time applicability.

### **5.3 Natural Language Processing for Geopolitical Intelligence**

Natural language processing (NLP) techniques enable extraction of geopolitical risk signals from unstructured text sources, transforming qualitative intelligence into quantitative indicators suitable for integration with digital twin systems (Voicu, 2025). Named entity recognition (NER) identifies relevant actors (countries, organizations, political figures), events (sanctions, conflicts, policy changes), and locations mentioned in news articles and policy documents. Sentiment

analysis and event extraction algorithms then assess the tone and implications of these mentions for supply chain operations.

Transformer-based language models, particularly BERT and GPT architectures, have revolutionized NLP capabilities for geopolitical risk assessment (Stanojevic, 2025). These models, pre-trained on massive text corpora and fine-tuned on domain-specific data, achieve state-of-the-art performance in tasks such as risk classification, impact assessment, and temporal forecasting of geopolitical events. However, the literature reveals limited attention to domain adaptation challenges when applying general-purpose language models to specialized supply chain and logistics contexts.

A critical gap identified in the NLP literature is the lack of standardized benchmarks and evaluation datasets for geopolitical risk extraction in supply chain contexts. Existing studies employ ad hoc evaluation approaches and proprietary datasets, hindering reproducibility and comparative assessment of different NLP techniques. The development of shared task datasets and evaluation protocols represents a priority for advancing this research stream.

#### **5.4 Hybrid and Ensemble Approaches**

Recognizing the complementary strengths and limitations of different AI methodologies, recent research increasingly emphasizes hybrid and ensemble approaches that combine multiple techniques (Mittal et al., 2025). For example, NLP-based geopolitical risk extraction can feed into supervised learning models for disruption prediction, which in turn inform RL-based adaptive decision-making within digital twin simulations. These integrated pipelines leverage the specialized capabilities of each methodology while mitigating individual weaknesses.

Ensemble methods that aggregate predictions from multiple models--such as stacking, boosting, and Bayesian model averaging--demonstrate improved robustness and accuracy compared to individual models (Nwokocha et al., 2025). Ensemble approaches prove particularly valuable for handling the deep uncertainty characteristic of geopolitical risks, as they provide probabilistic predictions that quantify confidence and capture alternative scenarios.

However, the literature reveals limited theoretical guidance on optimal ensemble design for geopolitical risk prediction. Key questions remain underexplored: How should models be weighted in ensemble predictions? How can ensemble diversity be balanced against individual model accuracy? How should ensembles adapt as new data becomes available and geopolitical contexts evolve? Addressing these questions requires integration of insights from machine learning theory, decision theory, and supply chain management.

## **6. Discussion**

### **6.1 Critical Assessment of Current State**

The synthesis of 142 academic papers and 20 industry reports reveals a field characterized by rapid technological advancement, growing practitioner interest, but persistent gaps in theoretical grounding, methodological rigor, and empirical validation. While digital twin technology and AI methodologies have matured significantly as individual domains, their integration for geopolitical risk simulation in supply chains remains nascent and fragmented. The literature exhibits several critical tensions and unresolved challenges that warrant careful examination.

First, a fundamental tension exists between model complexity and interpretability. The most sophisticated AI architectures--deep neural networks, ensemble methods, complex agent-based simulations--offer superior predictive accuracy but function as "black boxes" that provide limited insight into causal mechanisms (Sharma, 2024). For supply chain decision-makers facing high-stakes choices under geopolitical uncertainty, the lack of interpretability undermines trust and limits practical adoption. Explainable AI (XAI) techniques offer potential solutions, but their application to complex supply chain digital twins remains underexplored.

Second, the literature reveals significant methodological limitations in empirical validation. Many studies rely on historical backtesting or synthetic simulation scenarios rather than prospective validation under actual geopolitical disruptions (Katsaliaki et al., 2021). This validation gap is particularly problematic given the non-stationary nature of geopolitical risks, where historical patterns may provide poor guidance for future scenarios. The scarcity of longitudinal studies tracking digital twin performance across multiple disruption events limits confidence in claimed benefits.

Third, data integration challenges persist as a major barrier to practical implementation. The heterogeneity of data sources, formats, and quality standards across global supply chains complicates the creation of unified digital twin representations (Mittal et al., 2025). Cross-border data sharing faces regulatory barriers (e.g., GDPR, data localization requirements), commercial sensitivities, and technical interoperability issues. The literature offers limited guidance on pragmatic approaches to data integration that balance comprehensiveness with feasibility.

Fourth, the incorporation of exogenous geopolitical signals into predictive frameworks remains theoretically and methodologically underdeveloped. Most AI models focus on endogenous supply chain variables (inventory levels, lead times, demand patterns) while treating geopolitical risks as external shocks (Voicu, 2025). This approach fails to capture

the co-evolution of supply chain strategies and geopolitical landscapes, where firm decisions (e.g., reshoring, diversification) influence political dynamics, which in turn affect future supply chain options. Agent-based modeling and game-theoretic approaches offer potential frameworks for capturing these feedback loops but remain underutilized.

Fifth, the literature exhibits limited attention to organizational and human factors that mediate technology adoption and effectiveness. Digital twin systems do not operate in isolation but must integrate with existing organizational processes, decision-making structures, and human expertise (Rai et al., 2025). The sociotechnical challenges of implementation—including change management, skill development, and organizational culture—receive insufficient attention in the predominantly technology-focused literature.

## **6.2 Empirical Implementation Challenges in Developing Economies**

While the literature on AI-powered digital twin systems for geopolitical risk simulation has primarily focused on advanced economies and multinational corporations, the implementation challenges in developing economies present distinct and often more severe barriers that warrant explicit attention. These challenges span data infrastructure, institutional capacity, geopolitical exposure profiles, and resource constraints, fundamentally shaping the feasibility and design of digital twin deployments in emerging markets.

### **Data Infrastructure Limitations**

Developing economies face significant data infrastructure deficits that undermine the foundational requirements of digital twin systems. Unreliable internet connectivity, particularly in rural and peri-urban areas where manufacturing facilities and logistics hubs are often located, disrupts the real-time data synchronization essential for digital twin functionality (Nwokocha et al., 2025). The absence of comprehensive IoT sensor networks in warehouses, transportation fleets, and border crossings limits the granularity and timeliness of operational data. Many small and medium-sized enterprises (SMEs) in developing economies operate with fragmented or legacy enterprise resource planning (ERP) and warehouse management systems (WMS) that lack standardized data formats and application programming interfaces (APIs) necessary for seamless integration with digital twin platforms (Rahul et al., 2025).

Data quality and standardization present additional challenges. Inconsistent data entry practices, manual record-keeping, and limited data governance frameworks result in incomplete, inaccurate, or incompatible datasets that compromise the reliability of digital twin representations

(Onebunne et al., 2025). Cross-border data flows face regulatory fragmentation, with varying data localization requirements, privacy regulations, and customs documentation standards across jurisdictions. These infrastructure limitations necessitate digital twin architectures that can function effectively with sparse, noisy, and intermittent data—a design constraint rarely addressed in the literature focused on data-rich advanced economy contexts.

### **Institutional and Regulatory Constraints**

The institutional environment in many developing economies presents structural barriers to digital twin adoption. Weak regulatory frameworks for data sharing and intellectual property protection discourage firms from participating in collaborative digital twin ecosystems that require transparency across supply chain partners (Stanojevic, 2025). Limited public-private partnerships and data-sharing initiatives restrict access to critical geopolitical intelligence, customs data, and infrastructure status information that would enhance digital twin predictive capabilities.

Bureaucratic barriers to technology adoption, including complex import regulations for hardware and software, restrictive foreign investment rules for technology services, and lengthy approval processes for cloud computing infrastructure, slow implementation timelines and increase costs (Rahul et al., 2025). Corruption and informal practices in customs clearance, transportation permitting, and regulatory compliance introduce unpredictable variability that is difficult to model in digital twin simulations, as these informal dynamics operate outside documented processes and data systems.

The shortage of AI and data science talent in developing economies constrains both initial implementation and ongoing maintenance of digital twin systems. Universities and technical training institutions often lack curricula aligned with emerging technologies, while brain drain draws skilled professionals to advanced economies (Nwokocha et al., 2025). This talent gap necessitates reliance on external consultants or offshore development, increasing costs and creating dependencies that undermine long-term sustainability.

### **Geopolitical Exposure Differences**

Developing economies often face distinct and more volatile geopolitical risk profiles compared to advanced economies, requiring digital twin systems calibrated to different threat landscapes. Currency instability and exchange rate volatility introduce financial risks that compound operational disruptions, as supply chain costs denominated in foreign currencies fluctuate unpredictably (Stanojevic, 2025). Political instability, including regime changes, civil unrest, and weak rule of law, creates discontinuous risk patterns that challenge AI models



trained on more stable advanced economy contexts.

Border disputes and regional conflicts disproportionately affect developing economies, particularly those in geopolitically contested regions. Digital twin systems must account for sudden border closures, militarized zones, and infrastructure damage that rarely occur in advanced economies (Voicu, 2025). Dependence on a narrow set of trading partners or export commodities creates concentration risks that amplify the impact of bilateral geopolitical tensions, requiring digital twin scenario analysis to emphasize tail risks and extreme events rather than incremental disruptions.

Developing economies also face asymmetric power dynamics in global supply chains, often serving as suppliers to multinational corporations with limited bargaining power or visibility into downstream demand shifts driven by geopolitical factors in advanced economies. Digital twin systems in these contexts must address information asymmetries and power imbalances that shape risk exposure and mitigation options (Onebunne et al., 2025).

### **Financial and Resource Barriers**

The high upfront costs of digital twin deployment—including hardware infrastructure (IoT sensors, edge computing devices), software licenses, cloud computing services, and implementation consulting—present prohibitive barriers for resource-constrained firms in developing economies. Return on investment (ROI) uncertainty, particularly for SMEs with limited financial buffers, discourages adoption even when potential benefits are recognized (Rahul et al., 2025). Limited access to financing for technology investments, due to underdeveloped venture capital markets and risk-averse banking sectors, further constrains adoption.

Cloud infrastructure availability and affordability vary significantly across developing economies. While major cloud providers have expanded data center presence in some emerging markets, many regions lack local cloud infrastructure, requiring data transmission to distant data centers with associated latency, bandwidth costs, and data sovereignty concerns (Nwokocha et al., 2025). The total cost of ownership for digital twin systems, including ongoing subscription fees, data storage costs, and maintenance expenses, can exceed the financial capacity of smaller firms that constitute the majority of supply chain participants in developing economies.

### **Recommendations for Context-Adapted Implementation**

Addressing these challenges requires context-adapted digital twin architectures and implementation strategies tailored to developing economy constraints. Lightweight digital twin frameworks that prioritize essential functionality over comprehensive simulation can reduce computational

requirements and costs while delivering core benefits (Onebunne et al., 2025). Edge computing approaches that process data locally before transmitting aggregated insights to cloud platforms can mitigate bandwidth limitations and reduce latency. Modular, open-source digital twin platforms can lower licensing costs and enable customization to local contexts, though they require stronger technical capacity for implementation and maintenance.

Federated learning approaches offer promising solutions to data scarcity and privacy concerns by enabling AI model training across distributed datasets without centralizing sensitive information (Rahul et al., 2025). This approach allows firms to benefit from collective intelligence while maintaining data sovereignty and protecting competitive information. Transfer learning techniques that adapt AI models pre-trained on advanced economy data to developing economy contexts can accelerate deployment and reduce data requirements, though careful validation is necessary to ensure model relevance.

Capacity-building partnerships between multinational corporations, technology providers, universities, and government agencies can address talent shortages and institutional gaps. These partnerships might include training programs for supply chain professionals, collaborative research initiatives, and pilot projects that demonstrate feasibility and build local expertise (Nwokocha et al., 2025). Public-private data-sharing initiatives, potentially facilitated by international development organizations, can improve access to geopolitical intelligence and infrastructure data while establishing governance frameworks that protect commercial interests.

Phased rollout strategies that begin with limited pilot implementations in specific supply chain segments or geographic regions can reduce upfront costs, demonstrate value, and build organizational capabilities incrementally (Onebunne et al., 2025). These pilots should prioritize high-impact, low-complexity use cases—such as real-time shipment tracking or supplier risk monitoring—that deliver tangible benefits quickly and build momentum for broader adoption. Hybrid approaches that combine digital twin technology with existing manual processes and expert judgment can bridge capability gaps during transition periods.

Finally, international standards and interoperability protocols specifically designed for resource-constrained contexts can facilitate adoption by reducing integration complexity and enabling economies of scale across developing economies (Rahul et al., 2025). Development of regional digital twin platforms that serve multiple firms or countries can distribute costs and create network effects, though governance structures must carefully balance collaboration with competition

concerns.

The successful deployment of AI-powered digital twin systems for geopolitical risk simulation in developing economies requires not merely technology transfer but fundamental adaptation of architectures, methodologies, and implementation strategies to local constraints and opportunities. Future research should prioritize empirical studies in developing economy contexts, co-design approaches that engage local stakeholders, and evaluation frameworks that account for resource constraints and institutional realities. Only through such context-sensitive research and practice can the benefits of digital twin technology be equitably distributed across the global supply chain ecosystem.

### 6.3 Theoretical Contributions and Gaps

From a theoretical perspective, the integration of digital twin technology with AI-driven geopolitical risk simulation offers potential contributions to several scholarly domains. For supply chain management theory, digital twins operationalize dynamic capabilities by providing concrete mechanisms for sensing, seizing, and transforming in response to environmental turbulence (Rai et al., 2025). This operationalization enables more rigorous empirical testing of dynamic capabilities theory and clarifies the micro-foundations through which organizational agility is achieved.

For information systems research, AI-powered digital twins exemplify sociotechnical systems that blur boundaries between physical and digital, human and algorithmic decision-making. The co-evolution of digital twin systems and organizational practices raises fundamental questions about agency, control, and the nature of managerial work in algorithmically mediated environments (Sharma, 2024). These questions connect to broader debates about AI governance, algorithmic accountability, and the future of work.

For international business and political economy, the integration of geopolitical risk assessment with supply chain optimization challenges traditional assumptions about the separability of political and economic spheres. Digital twin systems that simultaneously optimize for cost efficiency and geopolitical resilience make explicit the trade-offs and complementarities between economic and political objectives, potentially informing theories of firm strategy under political uncertainty (Stanojevic, 2025).

However, significant theoretical gaps persist. The literature lacks comprehensive frameworks that integrate micro-level technological capabilities (digital twin architectures, AI algorithms) with meso-level organizational processes (decision-making, learning, adaptation) and macro-level institutional contexts (regulatory regimes, geopolitical

structures). Such multi-level frameworks are essential for understanding how digital twin systems generate value and under what conditions they succeed or fail.

Additionally, the temporal dynamics of digital twin effectiveness remain undertheorized. How do digital twin systems evolve and adapt as geopolitical contexts shift? How do organizations learn from digital twin-supported decisions and refine their systems over time? What are the path dependencies and lock-in effects associated with particular digital twin architectures? These questions require longitudinal theoretical frameworks that capture learning, adaptation, and co-evolution.

### 6.4 Practical Implications

For logistics practitioners and supply chain managers, the literature offers several actionable insights while also highlighting important caveats. First, digital twin implementation should be approached incrementally, beginning with well-defined use cases that offer clear value propositions and manageable complexity (Rai et al., 2025). Pilot projects focused on specific supply chain segments, geographic regions, or risk types enable learning and capability building while limiting downside risk.

Second, data integration and quality management must be prioritized as foundational capabilities. The value of digital twin systems depends critically on the availability, accuracy, and timeliness of underlying data (Mittal et al., 2025). Organizations should invest in data governance frameworks, standardized data formats, and integration platforms before pursuing sophisticated AI capabilities.

Third, human expertise and judgment remain essential complements to AI-powered digital twins. Rather than replacing human decision-makers, digital twin systems should be designed to augment human capabilities by providing enhanced situational awareness, scenario exploration tools, and decision support (Sharma, 2024). Organizational processes should explicitly define roles and responsibilities for human oversight, particularly for high-stakes decisions under deep uncertainty.

Fourth, collaboration and information sharing across supply chain partners enhance digital twin effectiveness but require careful governance. Establishing clear protocols for data sharing, intellectual property protection, and benefit distribution can enable collaborative digital twin ecosystems while protecting competitive interests (MDPI, 2025). Industry consortia and standardization efforts can facilitate interoperability and reduce implementation costs.

Fifth, organizations should maintain realistic expectations about digital twin capabilities and limitations. Digital twins are not crystal balls that eliminate uncertainty but rather tools that

enable more informed decision-making under uncertainty (Voicu, 2025). Overconfidence in model predictions or neglect of model limitations can lead to poor decisions and undermine trust in digital twin systems.

## 7. Future Research Directions

The synthesis of current literature reveals several critical priorities for future research that would advance both theoretical understanding and practical application of AI-powered digital twin systems for geopolitical risk simulation.

### 7.1 Methodological Priorities

Longitudinal empirical studies that track digital twin performance across multiple geopolitical disruption events are urgently needed to validate claimed benefits and identify success factors (Katsaliaki et al., 2021). These studies should employ rigorous quasi-experimental designs that compare outcomes for firms with and without digital twin systems, controlling for confounding factors such as firm size, industry, and geographic exposure. Prospective validation, where digital twin predictions are recorded before disruptions occur and subsequently evaluated, would provide stronger evidence than retrospective backtesting.

Comparative case studies examining digital twin implementations across different organizational contexts, industries, and geopolitical environments would illuminate contingency factors that moderate effectiveness (Rai et al., 2025). These studies should employ structured protocols for data collection and analysis to enable systematic cross-case comparison and theory building.

Experimental research using controlled simulations or field experiments could isolate the causal effects of specific digital twin features (e.g., real-time synchronization, AI-driven prediction, scenario simulation) on decision quality and supply chain performance (Sharma, 2024). Such experiments would complement observational studies by providing stronger causal inference while acknowledging limitations in external validity.

### 7.2 Technological Research Priorities

The development of standardized digital twin architectures and interoperability protocols represents a critical priority for enabling scalable, cost-effective implementations (Wasi et al., 2025). Research should address technical standards for data formats, API specifications, and integration frameworks that enable plug-and-play connectivity across heterogeneous systems. Open-source reference implementations could accelerate adoption and facilitate comparative research.

Explainable AI techniques tailored to supply chain digital twin applications require focused research attention. Methods for generating human-interpretable explanations of AI

predictions, identifying key drivers of risk assessments, and quantifying prediction uncertainty would enhance trust and practical utility (Sharma, 2024). Research should balance explanation fidelity (accuracy of explanations) with comprehensibility (ease of understanding by non-technical users).

Multi-modal risk integration frameworks that synthesize diverse geopolitical risk signals—including structured data (economic indicators, trade statistics), unstructured text (news, policy documents), and alternative data sources (satellite imagery, social media)—represent a promising research frontier (Voicu, 2025). Deep learning architectures capable of learning joint representations across modalities could enhance predictive accuracy and provide richer situational awareness.

Federated learning and privacy-preserving AI techniques offer potential solutions to data sharing barriers in supply chain digital twin ecosystems (Mittal et al., 2025). Research should address technical challenges in training accurate models on distributed, heterogeneous datasets while providing formal privacy guarantees. Applications to supply chain contexts with multiple competing firms and regulatory constraints present unique challenges.

### 7.3 Theoretical Research Priorities

Multi-level theoretical frameworks that integrate technological, organizational, and institutional perspectives on digital twin effectiveness represent a critical gap (Rai et al., 2025). Such frameworks should specify mechanisms linking digital twin capabilities to organizational outcomes, moderating factors at different levels of analysis, and feedback loops between levels. Empirical testing of these frameworks would advance understanding of when and how digital twins generate value.

Theories of human-AI collaboration in supply chain decision-making under uncertainty require development. How do human decision-makers integrate AI-generated insights with their own expertise and judgment? What cognitive biases or heuristics influence reliance on AI recommendations? How do trust, transparency, and explanation quality affect human-AI collaboration effectiveness? Addressing these questions requires integration of insights from cognitive psychology, decision sciences, and information systems research (Sharma, 2024).

Theories of digital twin evolution and adaptation over time would illuminate learning processes and path dependencies. How do organizations refine digital twin systems based on experience? What organizational learning mechanisms enable effective adaptation? How do initial architectural choices constrain or enable future evolution? Longitudinal theoretical frameworks informed by organizational learning theory and

evolutionary economics could address these questions.

#### 7.4 Contextual Research Priorities

Research examining digital twin applications in developing economies and emerging markets is critically needed, as current literature disproportionately focuses on advanced economies and large multinational corporations (Nwokocha et al., 2025). Studies should address context-specific challenges including data infrastructure limitations, institutional constraints, and resource scarcity, while identifying adapted implementation strategies suitable for these contexts.

Industry-specific research that accounts for unique characteristics of different sectors—including perishable goods logistics, pharmaceutical supply chains, automotive manufacturing, and electronics—would provide more actionable guidance (Katsaliaki et al., 2021). Geopolitical risk profiles, supply chain structures, and regulatory environments vary significantly across industries, necessitating tailored approaches.

Research on small and medium-sized enterprises (SMEs) and their participation in digital twin ecosystems represents another critical gap. Most literature focuses on large corporations with substantial resources and technical capabilities, yet SMEs constitute the majority of supply chain participants globally (Rahul et al., 2025). Understanding barriers to SME adoption and developing accessible, affordable digital twin solutions would enhance supply chain resilience more broadly.

### 8. Conclusion

This critical literature review has systematically examined the integration of AI-powered digital twin systems for geopolitical risk simulation in global logistics, synthesizing 142 academic papers and 20 industry reports published between 2019 and 2026. The review reveals a field characterized by rapid technological advancement, growing practitioner interest, but persistent gaps in theoretical grounding, methodological rigor, and empirical validation.

Key findings indicate that digital twin technology has matured significantly for supply chain applications, offering capabilities for real-time monitoring, predictive analytics, and scenario simulation. When augmented with AI methodologies—particularly reinforcement learning, deep learning, and natural language processing—digital twins enable sophisticated geopolitical risk assessment and adaptive decision-making. However, critical challenges persist in data integration, model interpretability, empirical validation, and organizational implementation.

The review identifies fundamental tensions between model complexity and interpretability, challenges in incorporating exogenous geopolitical signals into predictive frameworks, and

limitations in cross-border data harmonization. Methodological limitations in empirical validation, particularly the scarcity of longitudinal studies tracking digital twin performance across actual disruptions, constrain confidence in claimed benefits. Organizational and human factors that mediate technology adoption and effectiveness receive insufficient attention in the predominantly technology-focused literature.

From a theoretical perspective, the integration of digital twin technology with AI-driven geopolitical risk simulation offers potential contributions to supply chain management theory, information systems research, and international business scholarship. Digital twins operationalize dynamic capabilities, exemplify sociotechnical systems that blur boundaries between physical and digital, and make explicit trade-offs between economic and political objectives in supply chain strategy. However, comprehensive multi-level frameworks integrating technological, organizational, and institutional perspectives remain underdeveloped.

Future research priorities emphasize the need for longitudinal empirical studies that validate digital twin effectiveness under actual geopolitical disruptions, standardized architectures and interoperability protocols that enable scalable implementations, explainable AI techniques that enhance trust and interpretability, and multi-modal risk integration frameworks that synthesize diverse geopolitical signals. Contextual research examining digital twin applications in developing economies, different industries, and small and medium-sized enterprises would enhance the breadth and practical relevance of the knowledge base.

For logistics practitioners, the review offers actionable insights while highlighting important caveats. Digital twin implementation should be approached incrementally, with data integration and quality management prioritized as foundational capabilities. Human expertise and judgment remain essential complements to AI-powered systems, and realistic expectations about capabilities and limitations are critical for successful adoption. Collaboration and information sharing across supply chain partners can enhance effectiveness but require careful governance.

The convergence of digital twin technology, artificial intelligence, and geopolitical risk assessment represents a transformative paradigm in global logistics management. As geopolitical tensions continue to escalate and supply chain disruptions become more frequent and severe, the imperative for proactive, adaptive risk management capabilities intensifies. AI-powered digital twin systems offer promising tools for navigating this volatile landscape, but realizing their potential requires continued research, technological innovation, and organizational learning. This review provides a comprehensive



foundation for advancing both scholarly understanding and practical application of these emerging capabilities, while identifying critical gaps and priorities that should guide future efforts.

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